

Practice Quiz 05B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

Proof by contradiction with sets.

1. Suppose $A \subseteq B$ and $B \cap C = \emptyset$. Prove by contradiction that $A \cap C = \emptyset$.

Solution Suppose for contradiction that $A \cap C \neq \emptyset$. Then there is an element $x \in A \cap C$, so $x \in A$ and $x \in C$. Because $A \subseteq B$, we also have $x \in B$. Thus $x \in B \cap C$, contradicting $B \cap C = \emptyset$. Therefore $A \cap C = \emptyset$. ■